MOLDOVA—DEBT SUSTAINABILITY ANALYSIS

A. Background

1. Moldova's total external debt at end-2005 is estimated at US$1.6 billion or 55 percent of GDP, down from 64 percent in 2004.[22] The public and publicly-guaranteed (PPG) external debt stock was reduced from 35 percent of GDP to 28 percent. It declined in dollar terms as well, partly because of appreciation of the US dollar with respect to other currencies (SDR, Euro, Yen), but also because of net repayments to creditors and cancellation of state guarantees on some of EBRD loans.[23] The NPV of external debt-to-exports is estimated to have fallen from 114 percent to 101 percent. Debt service indicators have also improved with the PPG debt service-to-exports ratio declining from 9 percent to 7 percent. Domestic currency public debt is modest at about 10 percent of GDP (end–2005).

B. External Debt Sustainability

2. The baseline projection shows further strengthening of the debt position throughout the period. The NPV of external debt-to-GDP ratio and debt-to-exports is expected to stabilize at about 8 percent and 16 percent, respectively, by 2025. The debt-service ratio is also projected to continue to decline and remain well within the debt burden thresholds as defined by the World Bank's CPIA-based classification.[24] The alternative scenarios and stress tests do not significantly alter the medium and long-term path of external debt dynamics indicating the robustness of the conclusions.

C. Public Debt Sustainability

3. The ratio of public debt to GDP has also declined significantly in recent years. Public debt is estimated to be 39 percent at end-2005, compared to 74.8 percent in 2002, reflecting strong growth and real exchange rate appreciation. Consistent with the authorities' medium term fiscal program, the baseline scenario projects a further decline in the debt-to-GDP ratio, with the NPV of public sector debt-to-GDP ratio declining to 10 percent by 2025. The alternative scenarios and stress tests also show declining ratios over time, with the exception of the alternative scenario with permanently lower GDP growth (2.7 percent long-run growth vs. 4 percent in the baseline) and the stress test with GDP growth at its 10-year historical average minus one standard deviation in 2006–07 (implying a growth rate of

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[22] This consists of $821 million in public and publicly guaranteed debt (28 percent of GDP) including $56.3 million in arrears (1.9 percent of GDP), $488 million in private debt (17 percent of GDP) and $286.6 million in energy arrears (10 percent of GDP). Private debt and energy arrears are estimates of the authorities.

[23] EBRD loans to Giurgiulesti Oil Terminal and Vininvest have been transferred to private investors in debt-for-equity swap transactions, thereby relieving the government of any obligations to the EBRD on these loans.

[24] According to the classification, Moldova is among poor performers, implying the following critical thresholds: NPV/exports = 100 percent, NPV/GDP = 30 percent, NPV/revenues = 200 percent, debt service/exports = 15 percent, debt service/revenues = 25 percent.

-3.4 percent for these two years). The results based on historical averages are not good proxies for expected future developments in Moldova, as this period covers the transition period as well as the 1998 regional financial crisis. Further, since the DSAs are static exercises, the results assume that spending would not adjust to this lower growth. However, the alternative scenario with permanently lower GDP growth emphasizes the importance for Moldova to continue to improve future growth prospects in order to maintain public debt sustainability.

D. Conclusions

4. Rather than a structural debt sustainability problem, Moldova faces short-term liquidity difficulties, which stem from the projected financing gaps and the stock of arrears owed to bilateral creditors, chiefly to Russia, the U.S. and Japan. Reengagement of IFI's and the international donor community with fresh concessional financing and possibly rescheduling of its external debt, including arrears, would alleviate the liquidity problem and free up resources for other development needs.

Table 1a. Moldova: External Debt Sustainability Framework, Baseline Scenario, 2003–25 1/
(In percent of GDP, unless otherwise indicated)

	Actuals		Estimate	Historical Average 6/	Standard Deviation 6/	Projections									
	2003	2004	2005			2006	2007	2008	2009	2010	2011	2006-11 Average	2015	2025	2012-25 Average
External debt (nominal) 1/	**88.7**	**63.8**	**54.7**			**51.3**	**49.7**	**47.3**	**44.2**	**41.0**	**37.8**		**28.7**	**24.6**	
of which public and publicly guaranteed (PPG)	52.5	35.2	28.2			24.7	22.9	21.4	19.9	18.6	17.4		13.7	10.5	
Change in external debt	-12.2	-24.9	-9.1			-3.4	-1.7	-2.3	-3.2	-3.2	-3.2		-1.4	0.2	
Identified net debt-creating flows	-13.4	-21.3	-7.4			-4.1	-3.3	-3.5	-3.4	-3.6	-4.2		-2.8	-0.3	
Non-interest current account deficit	**4.3**	**0.4**	**3.2**	**2.5**	**2.8**	**3.3**	**3.4**	**3.5**	**3.7**	**3.5**	**3.4**	**3.5**	**3.6**	**3.5**	3.6
Deficit in balance of goods and services	33.8	30.5	39.8			42.7	42.0	41.0	39.9	38.9	38.2		37.7	37.7	
Exports	53.5	51.2	52.6			53.7	54.0	53.3	53.0	52.7	52.6		52.7	52.7	
Imports	87.2	81.7	92.4			96.4	96.0	94.3	92.9	91.6	90.8		90.4	90.4	
Net current transfers (negative = inflow)	-15.4	-14.0	-19.0	-14.2	3.8	-20.4	-20.0	-19.4	-18.9	-18.6	-18.4	-19.3	-18.3	-18.3	-18.3
Other current account flows (negative = net inflow)	-14.1	-16.1	-17.6			-19.0	-18.7	-18.1	-17.3	-16.8	-16.4		-15.8	-15.9	
Net FDI (negative = inflow)	**-3.9**	**-3.0**	**-6.0**	**-5.4**	**2.7**	**-6.5**	**-6.5**	**-6.7**	**-6.7**	**-6.7**	**-7.2**	**-6.7**	**-6.4**	**-4.1**	-5.6
Endogenous debt dynamics 2/	**-13.7**	**-18.7**	**-4.5**			**-0.8**	**-0.2**	**-0.3**	**-0.3**	**-0.3**	**-0.4**		**0.0**	**0.3**	
Contribution from nominal interest rate	2.5	2.3	2.4			2.1	2.1	1.9	1.8	1.7	1.5		1.1	1.2	
Contribution from real GDP growth	-5.6	-5.0	-4.0			-2.9	-2.3	-2.2	-2.1	-2.0	-1.9		-1.1	-0.9	
Contribution from price and exchange rate changes	-10.6	-16.1	-2.9			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**1.2**	**-3.5**	**-1.8**			**0.7**	**1.7**	**1.1**	**0.2**	**0.4**	**0.9**		**1.4**	**0.5**	
o/w exceptional financing	-2.6	-1.0	-0.7			1.7	0.0	0.0	0.0	-2.5	0.0		0.0	0.0	
NPV of external debt 4/	...	58.6	52.9			49.0	46.8	44.3	41.2	38.1	35.1		26.0	22.6	
In percent of exports	...	114.4	100.5			91.1	86.7	83.1	77.7	72.3	66.6		49.4	42.9	
NPV of PPG external debt	**...**	**30.0**	**26.3**			**22.3**	**20.1**	**18.4**	**16.9**	**15.7**	**14.7**		**11.0**	**8.5**	
In percent of exports	...	58.5	50.0			41.6	37.2	34.6	31.9	29.8	28.0		20.9	16.1	
Debt service-to-exports ratio (in percent)	19.8	21.3	20.2			14.5	12.9	12.8	13.2	12.5	12.0		8.9	8.8	
PPG debt service-to-exports ratio (in percent)	12.5	9.3	6.9			5.5	4.6	4.1	3.6	2.7	2.3		1.9	1.6	
Total gross financing need (billions of U.S. dollars)	217.3	216.2	227.8			151.6	142.6	154.0	184.7	172.8	143.4		150.9	715.5	
Non-interest current account deficit that stabilizes debt ratio	16.5	25.2	12.3			6.7	5.0	5.9	6.8	6.6	6.6		5.0	3.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.6	7.4	7.1	2.3	5.5	6.0	5.0	5.0	5.0	5.0	5.0	5.2	4.0	4.0	4.1
GDP deflator in US dollar terms (change in percent)	11.8	22.1	4.8	9.8	6.6	7.3	7.0	7.1	6.4	5.3	4.9	6.3	4.9	3.0	4.2
Effective interest rate (percent) 5/	2.9	3.5	4.3	3.7	0.5	4.3	4.6	4.3	4.2	4.1	3.9	4.2	4.0	5.3	4.5
Growth of exports of G&S (US dollar terms, in percent)	20.9	25.6	15.3	16.8	7.0	16.2	13.0	11.0	11.0	10.0	10.0	11.9	9.0	7.1	8.5
Growth of imports of G&S (US dollar terms, in percent)	33.5	22.8	27.0	22.9	7.3	18.7	11.9	10.5	10.0	9.1	9.2	11.6	9.0	7.1	8.4
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	28.2	37.8	30.0	31.4	34.9	14.6	29.5	14.6	14.6	14.6
Memorandum item:															
Nominal GDP (billions of U.S. dollars)	1980.6	2597.9	2917.2			3318.7	3728.2	4192.0	4685.0	5179.1	5706.4		8125.7	17777.7	

Source: Fund staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

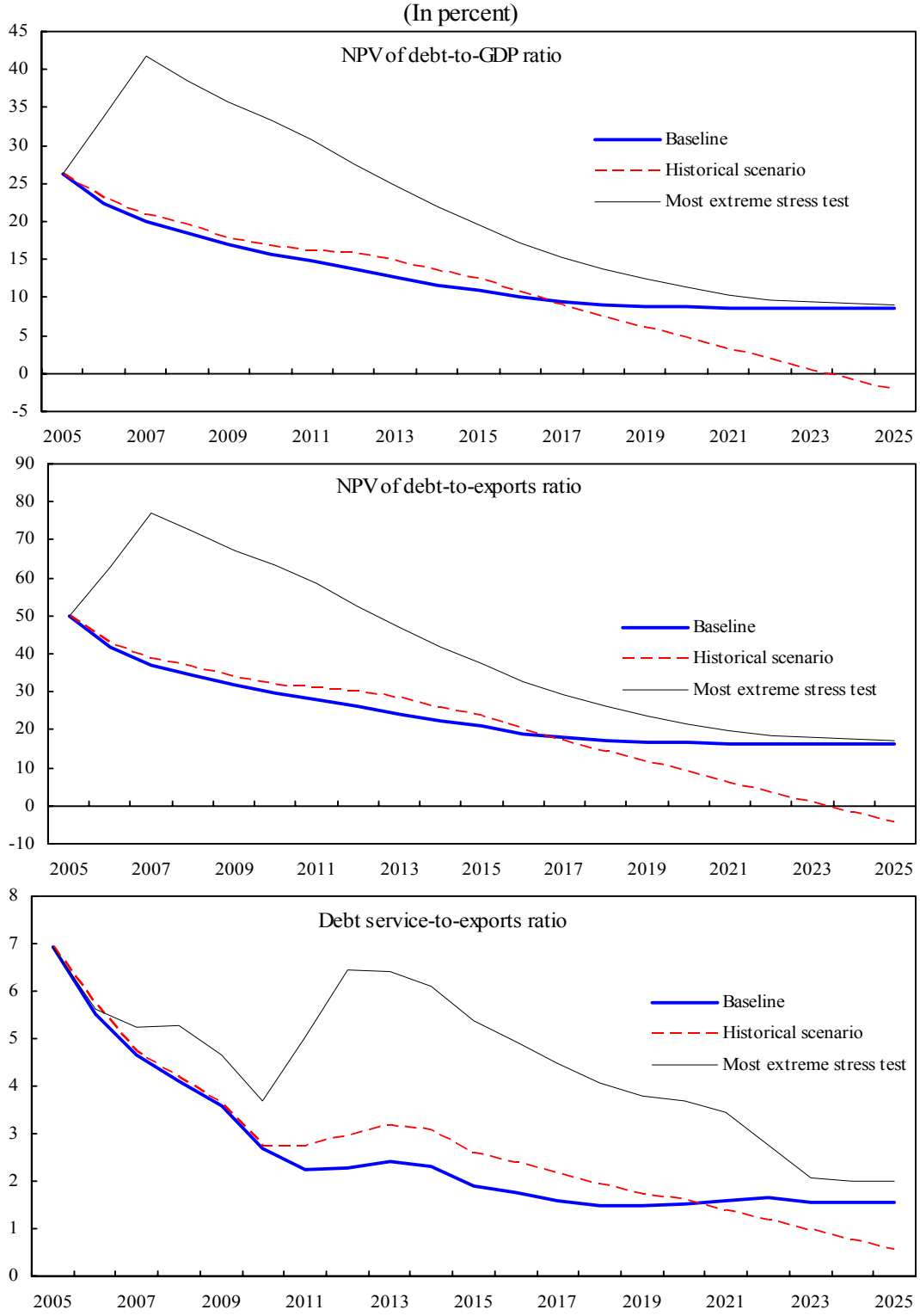
Table 1b. Moldova: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-25
(In percent)

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of debt-to-GDP ratio								
Baseline	26	22	20	18	17	16	11	9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	26	23	21	19	18	17	12	-2
A2. New public sector loans on less favorable terms in 2006-25 2/	26	22	21	19	18	17	13	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	26	24	24	22	20	19	13	10
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	26	25	26	27	25	24	15	9
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2006-07	26	23	22	20	18	17	12	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	26	34	42	39	36	33	20	9
B5. Combination of B1-B4 using one-half standard deviation shocks	26	28	37	35	33	30	19	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	26	31	28	25	23	21	15	12
NPV of debt-to-exports ratio								
Baseline	50	42	37	35	32	30	21	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	50	43	39	37	34	32	23	-4
A2. New public sector loans on less favorable terms in 2006-25 2/	50	42	38	36	34	33	25	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	50	45	37	34	32	30	21	16
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	50	49	52	56	52	49	31	18
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2006-07	50	43	37	35	32	30	21	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	50	63	77	72	67	63	37	17
B5. Combination of B1-B4 using one-half standard deviation shocks	50	51	61	59	55	52	32	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	50	57	37	34	32	29	21	16
Debt service ratio								
Baseline	7	6	5	4	4	3	2	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	7	6	5	4	4	3	3	1
A2. New public sector loans on less favorable terms in 2006-25 2/	7	6	5	4	4	3	2	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	7	6	5	4	4	3	3	2
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	7	6	5	5	4	3	4	2
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2006-07	7	6	5	4	4	3	3	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	7	6	5	5	5	4	5	2
B5. Combination of B1-B4 using one-half standard deviation shocks	7	6	5	5	4	3	4	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	7	8	5	4	4	3	3	2
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	18	18	18	18	18	18	**18**	18

Source: Fund staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1. Moldova: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2005–25
(In percent)



Source: Fund staff projections and simulations.

Table 2a. Moldova: Public Sector Debt Sustainability Framework, Baseline Scenario, 2002–25
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2002	2003	2004	Historical Average 5/	Standard Deviation 5/	2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025	2011-25 Average
Public sector debt 1/	74.8	60.3	47.2			39.0	33.5	30.4	27.7	25.2	23.1		15.9	11.8	
of which foreign-currency denominated	62.2	49.8	35.6			28.7	24.7	23.0	21.4	19.9	18.6		13.7	10.4	
Change in public sector debt	-5.3	-14.4	-13.1			-8.2	-5.5	-3.1	-2.8	-2.5	-2.1		-1.3	-0.2	
Identified debt-creating flows	-8.0	-16.8	-11.6			-6.8	-4.6	-3.3	-3.0	-2.4	-1.9		-0.7	-0.2	
Primary deficit	-0.4	-2.8	-2.6	0.4	3.6	-2.9	-0.6	-0.9	-0.6	-0.3	-0.2	-0.9	0.2	0.3	0.2
Revenue and grants	29.8	34.0	35.4			39.5	38.4	38.3	37.7	37.7	37.8		38.4	38.4	
of which : grants	0.6	0.3	0.4			1.3	0.5	0.6	0.2	0.3	0.2		0.2	0.2	
Primary (noninterest) expenditure	29.4	31.2	32.7			36.6	37.7	37.4	37.1	37.3	37.6		38.6	38.7	
Automatic debt dynamics	-7.0	-13.9	-8.9			-3.9	-3.9	-2.3	-2.4	-2.0	-1.7		-1.0	-0.5	
Contribution from interest rate/growth differential	-5.6	-5.0	-4.1			-3.4	-2.1	-1.3	-1.3	-1.2	-1.1		-0.6	-0.3	
of which : contribution from average real interest rate	0.2	-0.4	0.1			-0.3	0.1	0.3	0.2	0.1	0.1		0.1	0.1	
of which : contribution from real GDP growth	-5.8	-4.6	-4.2			-3.1	-2.2	-1.6	-1.4	-1.3	-1.2		-0.7	-0.5	
Contribution from real exchange rate depreciation	-1.4	-8.9	-4.8			-0.5	-1.8	-1.0	-1.1	-0.8	-0.6		...	...	
Other identified debt-creating flows	-0.6	-0.2	-0.1			-0.1	-0.1	-0.1	-0.1	-0.1	-0.1		0.0	0.0	
Privatization receipts (negative)	-0.6	-0.2	-0.1			-0.1	-0.1	-0.1	-0.1	-0.1	-0.1		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g., bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.6	2.4	-1.5			-1.4	-0.9	0.3	0.3	0.0	-0.2		-0.5	0.0	
NPV of public sector debt	64.5	49.9	40.0			34.7	26.4	22.9	19.0	16.6	14.8		11.2	9.5	
of which foreign-currency denominated	52.0	39.3	28.4			24.4	17.6	15.5	12.8	11.4	10.4		9.0	8.2	
of which external	52.0	39.3	28.4			24.4	17.6	15.5	12.8	11.4	10.4		9.0	8.2	
Gross financing need 2/	1.4	-0.2	-0.8			0.4	6.8	6.0	5.6	5.2	4.7		3.4	2.6	
NPV of public sector debt-to-revenue ratio (in percent) 3/	216.8	146.9	113.0			87.9	68.8	59.7	50.5	44.1	39.2		29.2	24.8	
of which external	174.8	115.8	80.3			61.8	45.9	40.4	34.0	30.2	27.4		23.4	21.3	
Debt service-to-revenue ratio (in percent) 3/ 4/	7.2	6.2	5.4			8.6	9.8	8.0	6.9	6.1	5.0		2.9	2.5	
Primary deficit that stabilizes the debt-to-GDP ratio	5.0	11.7	10.5			5.4	4.9	2.2	2.2	2.1	1.9		1.5	0.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.8	6.6	7.4	1.4	5.5	7.1	6.0	5.0	5.0	5.0	5.0	5.5	4.0	4.0	4.1
Average nominal interest rate on forex debt (in percent)	3.8	3.2	2.5	3.4	0.7	3.0	2.9	3.0	2.6	2.4	2.2	2.7	2.1	2.8	2.4
Average real interest rate on domestic currency debt (in percent)	-9.0	-8.9	1.1	-2.5	5.9	-3.4	-0.6	0.7	1.0	1.0	1.1	0.0	2.4	4.8	3.9
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.2	-15.0	-10.3	-7.4	6.8	-1.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	9.8	14.9	8.0	18.0	11.0	7.1	9.3	8.8	7.2	6.1	5.1	7.3	4.9	4.0	4.3
Growth of real primary spending (deflated by GDP deflator, in percent)	25.5	13.3	12.6	0.4	16.8	19.9	9.1	4.2	4.1	5.7	5.9	8.1	4.2	4.0	4.3

Sources: Moldovan authorities; and Fund staff estimates and projections.

1/ Includes general government gross debt.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues including grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

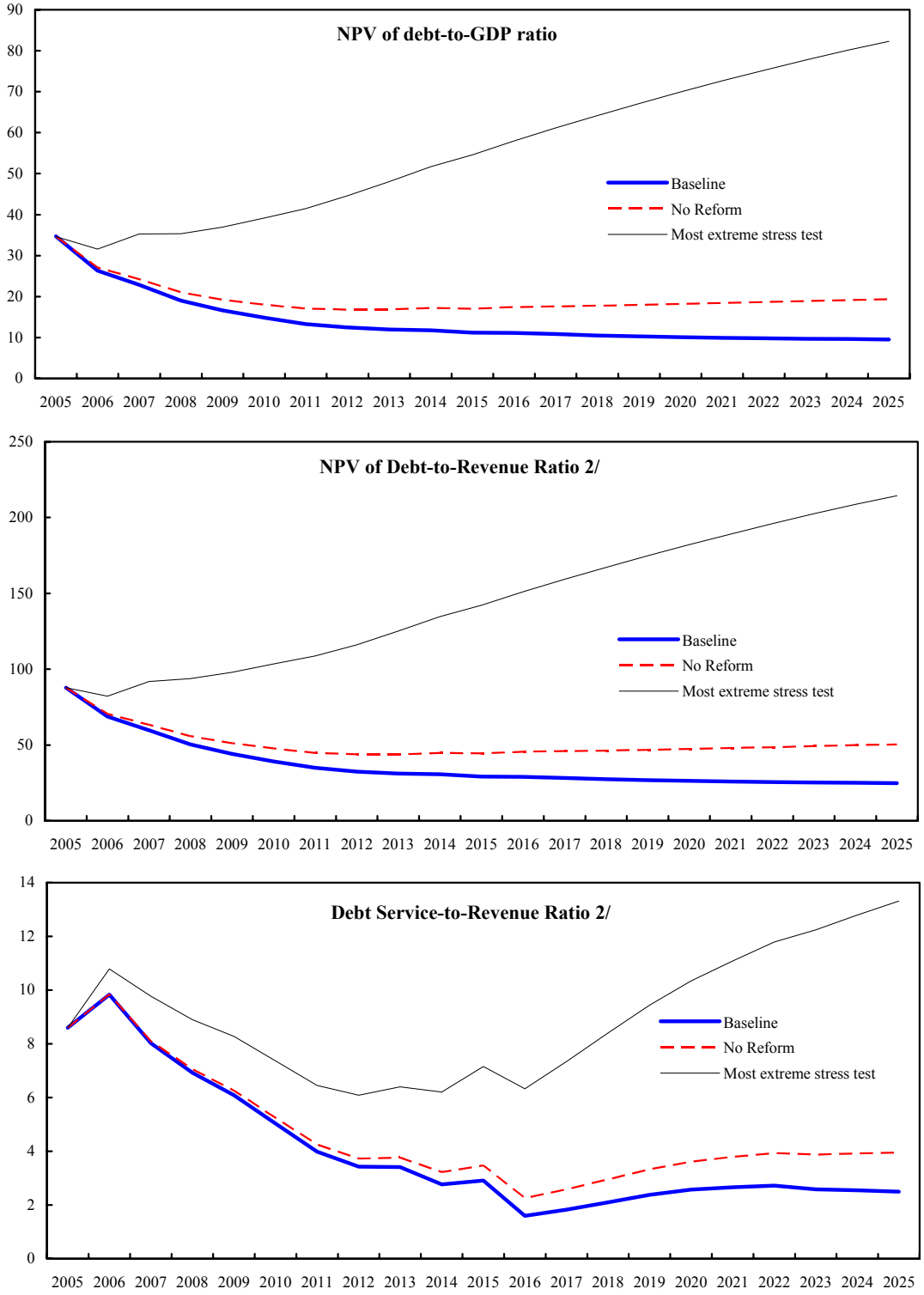
Table 2b. Moldova: Sensitivity Analysis for Key Indicators of Public Debt 2005–25

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of debt-to-GDP ratio								
Baseline	35	26	23	19	17	15	11	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	35	28	25	22	20	19	15	11
A2. Lower primary balance 1/	35	27	24	21	19	18	17	19
A3. Permanently lower GDP growth 2/	35	27	24	22	21	21	30	78
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-07	35	32	35	35	37	39	55	82
B2. Primary balance is at historical average minus one standard deviations in 2006-07	35	30	29	25	22	20	15	12
B3. Combination of B1-B2 using one half standard deviation shocks	35	30	30	25	22	20	15	12
B4. One-time 30 percent real depreciation in 2006	35	34	30	24	21	19	15	12
B5. 10 percent of GDP increase in other debt-creating flows in 2006	35	34	30	25	23	20	15	12
NPV of debt-to-revenue ratio 3/								
Baseline	88	69	60	50	44	39	29	25
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	88	72	66	59	53	49	38	30
A2. Lower primary balance 1/	88	71	63	56	51	48	44	50
A3. Permanently lower GDP growth 2/	88	70	64	57	55	54	78	202
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-07	88	82	92	94	98	103	142	214
B2. Primary balance is at historical average minus one standard deviations in 2006-07	88	77	76	66	58	53	39	31
B3. Combination of B1-B2 using one half standard deviation shocks	88	78	77	66	58	52	39	30
B4. One-time 30 percent real depreciation in 2006	88	90	78	65	57	50	38	31
B5. 10 percent of GDP increase in other debt-creating flows in 2006	88	88	78	67	60	54	40	31
Debt service-to-revenue ratio 3/								
Baseline	9	10	8	7	6	5	3	2
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	9	10	9	8	7	6	4	4
A2. Lower primary balance 1/	9	10	8	7	6	5	3	4
A3. Permanently lower GDP growth 2/	9	10	8	7	7	6	5	11
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-07	9	11	10	9	8	7	7	13
B2. Primary balance is at historical average minus one standard deviations in 2006-07	9	10	8	7	7	5	4	3
B3. Combination of B1-B2 using one half standard deviation shocks	9	10	9	8	7	6	4	3
B4. One-time 30 percent real depreciation in 2006	9	10	9	8	7	6	3	3
B5. 10 percent of GDP increase in other debt-creating flows in 2006	9	10	9	7	7	5	4	3

Sources: Moldovan authorities; and Fund staff estimates and projections.

1/ Assumes that the primary balance is at baseline minus 1 percentage points of GDP starting in 2006.
2/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
3/ Revenues are defined inclusive of grants.

Figure 2. Moldova: Indicators of Public Debt Under Alternative Scenarios, 2005–25 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2015.
2/ Revenue including grants.